

ICE: (Assume 30mpg)

Fuel used = 10,000 miles ÷ 30 mpg = 333.3 gallons

Fuel cost = 333.3 gal × \$3.42/gal ≈ **\$1,140**

Emissions= 333.3 gal x 19.5 pounds co2 per gal= roughly 6500 pounds of co2

Hybrid: (assume 52 mpg, low end for prius)

Fuel used = 10,000 miles ÷ 52 mpg ≈ 192.3 gallons

Fuel cost = 192.3 gal × \$3.42/gal ≈ **\$658**

Emissions= 192.3 gal x 19.5 pounds co2 per gal= 3749.85 pounds co2

EV: (assume 4 miles per kWh)

Electricity used = 10,000 miles ÷ 4 miles/kWh = 2,500 kWh

Electricity cost = 2,500 kWh × \$0.14/kWh = **\$350**

0.81 pounds co2 per kWh produced x 2500 kWh = 2025 pounds co2

You save about \$482 per 10000 miles by switch from ice to hybrid

You save about \$790 per 10000 miles by switching from ice to EV

IMPORTANT: YOU CAN DRIVE BOTH THE HYBRID AND EV 10000 miles (20000 miles total) and still save \$132 and reduce emissions by 725.1 pounds co2 for double the distance.